

JOHN KENNEDY KEVIN MANICKAM

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EDUCATION- BACHELOR OF COMPUTER AND DATA ENGINEERING

University College London- UCL (Exchange semester) [QS World Ranking: 9]

Jan 2026 - Jun 2026

- Recipient of the highly competitive Lee Shau Kee **Exchange Scholarship** 2025-26 (**HK\$48,000**)
- Selected as the **sole nominee from CityU's** College of Engineering for UCL's competitive exchange programme
- Will undertake upper-year modules in Machine Learning, Database Systems, Algorithms, and Programming

City University of Hong Kong [QS World Ranking: 63]

Sep 2023 - Jun 2027

- Current **CGPA: 3.9/4.3**
- Received **Dean's List recognition for five of five semesters.**
- Received an **Entrance Scholarship** valued at **HKD 160,000** for an exceptional academic record.
- Chan Sui Hung **Best Student Award 2025**, City University of Hong Kong

St Josephs Boys' High School, Bengaluru, India

Jun 2011 - Mar 2023

- 93.4% in Class 12 Board Exams (2023) (**Top 1%** of students)
- 94.6% in Class 10 Board Exams (2021) (**Top 1%** of students)

WORK EXPERIENCE

Software Developer Intern, Expando World Ltd.

Jun 2025 - Aug 2025

- Built and optimized cross-platform front-end architecture with **Next.js, React, TypeScript, TailwindCSS**, improving Time-to-Interactive, reducing code bundle size, and achieving an improvement in mobile responsiveness by **150ms** across **5+ device profiles**.
- Collaborated on **bug fixing, codebase maintenance, and UI enhancements** in a global development team, contributing to software stability and improved user experience through **Git-based workflows** and merge requests.
- Streamlined development and deployment by integrating **Supabase, Stripe CLI, and Docker**, and reducing environment setup time. Produced documentation that helped with onboarding queries and translations.

Junior Programmer Intern, Radiance Tech International Ltd.

Jun 2024 - Aug 2024

- Processed, validated, and **optimized 300+ 3D asset files** using Blender, Python, and Node.js scripts, reducing asset load errors by 35%, cutting manual QA time significantly, and improving retrieval speed.
- Enhanced internal development workflows by automating tasks with Bash, NPM scripts, and Docker, improving build stability and lowering repetitive manual steps by 20%. **Collaborated through Git, Lark, and Notion**, enabling a significantly faster sprint turnaround.
- Acquired hands-on experience in **Web Development and JavaScript**, worked primarily with the **three.js libraries**, applying them to support software tasks and build foundational skills in full-stack development.

PROJECTS

CART Decision Tree and SVM Classifier Engine | *Machine Learning, Python, NumPy, SciPy, Jupyter, Decision Trees, SVM*

[link](#)

- Implemented **CART decision tree** and **support vector machine (SVM)** algorithms from scratch in Jupyter notebooks, completing core functions for splitting, impurity evaluation, recursive tree construction, optimization, and prediction.
- Built and tested models using **NumPy/SciPy-based numerical operations** and notebook unit tests; validated correctness across local sanity checks and autograder-style workflows while preserving required function signatures and notebook structure.

Smart Pet Home - Project | *Arduino, C/C++, Arduino IoT Cloud, Power Electronics, Supercapacitors, Wi-Fi Embedded Systems*

[link](#)

- Designed and **prototyped a dynamo-powered running wheel** generating up to 30 V from bush baby motion; implemented rectification, supercapacitor storage, and voltage regulation to **autonomously drive a UVC sterilization lamp**.
- Single-handedly **enabled wireless IoT connectivity** by sourcing Wi-Fi-enabled Arduino boards and integrating with **Arduino IoT Cloud**; delivered a real-time mobile dashboard with remote control and live sensor monitoring.
- Led multiple **sensor integrations and historical trend visualization** into the IoT app, establishing the foundation for **automated climate control, light control, and automated feeder** in the smart bush baby habitat.

Morgan Stanley Code to Give Hackathon (Finalist) | *React Native, FastAPI, Supabase, JWT, Hugging Face, Docker*

[link](#)

- Developed a **full-stack education AI superapp** with gamified token economy, community forums, homework scanning, and analytics dashboards to boost parental engagement for underprivileged families.
- Contributed to **full-stack prototype** using FastAPI, Supabase, Redis, and Docker; team-integrated Hugging Face Qwen2.5-VL-7B for multilingual AI assistant and vision-language features, delivering a **production-ready solution**.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, JavaScript, TypeScript, HTML/CSS, Bash, Assembly, R, MATLAB

Frameworks & Libraries: React, React Native, Next.js, FastAPI, Flask, Node.js, TailwindCSS, Spring Boot, Bootstrap, Three.js

Developer Tools: Git, Docker, Supabase, PostgreSQL, MongoDB, Arduino IoT Cloud, VS Code, Visual Studio, npm/Yarn

Analytical Skills: Machine Learning, Statistical Analysis, Data Modelling & Visualization, Business Statistics, A/B Testing Fundamentals, Data Cleaning & ETL, Hypothesis Testing, Regression Analysis, Microsoft Office Suite (PowerPoint, Excel, Outlook, Access)

Hardware & Embedded Tools: Microcontrollers (Arduino, ESP32), IoT Prototyping, Sensor Integration, Power Electronics, Circuit Assembly, HDL (Verilog/VHDL), Water Pumps & Mechanical Automation, Energy Harvesting (Dynamo + Supercapacitors)

LEADERSHIP

Finance Tutor, Hall Management Team, City University of Hong Kong

Jun 2025 - Present

- **Managing finances**, reimbursements, and accounting while **enhancing administration** and fostering a welcoming, inclusive atmosphere within the residential halls at CityU.

President, Indian Student Association (ISA CityU)

Feb 2024 - Present

- Increased participant engagement by 30%, established flagship cross-university collaboration events, and oversaw the day-to-day management and operations of a cabinet that **served 200+ students and alumni**.

Organising Committee, St Josephs Boys' High School

Jun 2021 - Mar 2023

- Played a pivotal role in successfully **leading over 5 national and international festivals** during my last 3 years in school, driving impactful experiences and fostering collaboration.